BMS College of Engineering

(Autonomous College under VTU)

Department of Artificial Intelligence and Data Science

Project Work Bio-Inspired Algorithms (23DS7BSBIO) Final Presentation on

SwarmFolio: Multi-Objective Stock Portfolio Optimization Using Particle Swarm Optimization and Genetic Algorithms

Submitted by

Vivek Boora	1BM23AD073
Ayush Pandey	1BM23AD074
Nitin	1BM24AD404

Under the guidance of

Prof. Suman M
Assistant Professor
Dept of AI & DS

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Circle shows the slide number

Abstract

Three goals

- Maximise return
- Minimise volatility
- Minimise CVaR (tail loss)

under cap, floor and holding limits

Two optimisers

- Particle swarm
  MOPSO
- Genetic algorithm
  NSGA-II

One shared repair operator

What we found

0.70 vs 0.25

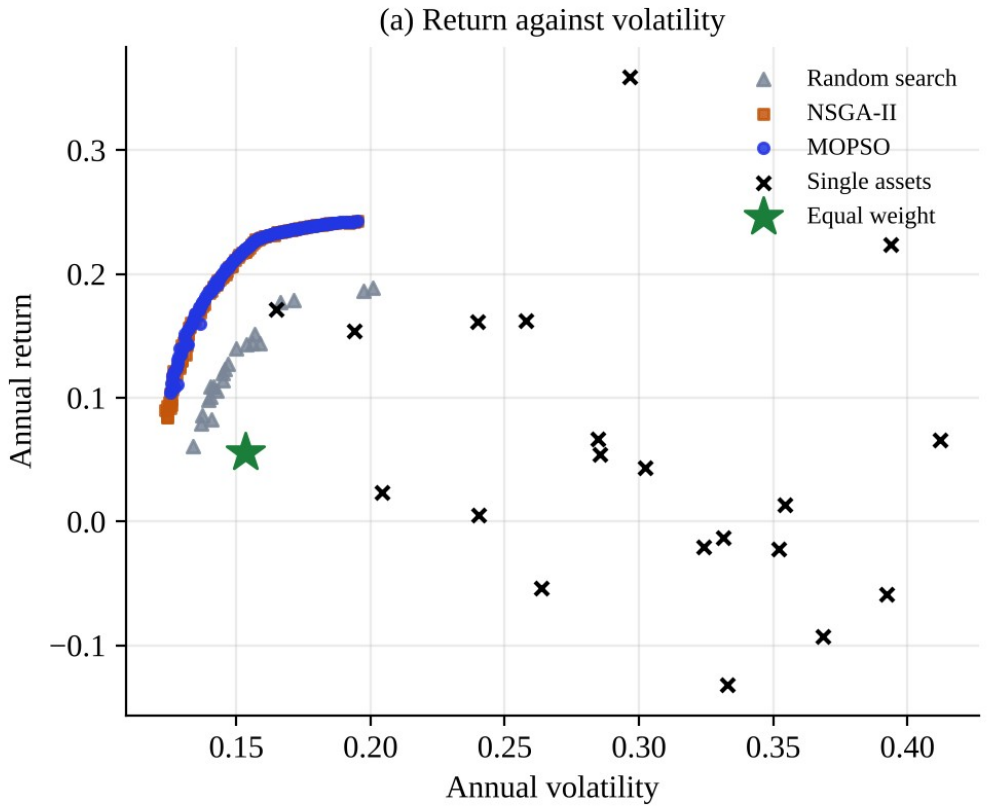
front quality, optimisers vs random search

1.34 to 0.53

Sharpe ratio, training to held-out data

A good front is not a promise of future return.

Introduction: the problem



Each dot is a feasible portfolio; the front is the best trade-off

Maximise annual return

Minimise volatility

Minimise CVaR: average loss on the worst 5% of days

Constraints applied to every portfolio

- Long-only
- Fully invested
- Cap 30% per stock
- Min holding 1%
- At most K holdings

Introduction: what we built

P

Python library

Portfolio model, repair operator, automated tests

T

Two optimisers

MOPSO and NSGA-II on one feasible space

F

Fair evaluation

Hypervolume, random search, 30 held-out markets

W

Web workbench

Watch the search and inspect any portfolio

Sustainable Development Goals

SDG 4 Quality Education

SDG 8 Decent Work and Economic Growth

SDG 9 Industry, Innovation, Infrastructure

Scope: simulated markets, long-only portfolios, no trading costs in the optimisation

Literature Survey

Paper Title	Name of Journal / Conference	Published Year	Novelty / Techniques	Limitations	Gap Identified
Portfolio selection (Markowitz)	The Journal of Finance	1952	Mean-variance model, efficient frontier	Variance only; no holding limits	Needs tail risk and constraints
Optimization of conditional value-at-risk (Rockafellar, Uryasev)	The Journal of Risk	2000	CVaR as a tail-risk measure	One risk measure at a time	Combine with return and volatility
Heuristics for cardinality constrained portfolio optimisation (Chang et al.)	Computers & Operations Research	2000	GA, tabu search, annealing with holding limits	Limits make the frontier hard to compute exactly	Fair comparison of heuristics
A fast and elitist multiobjective GA: NSGA-II (Deb et al.)	IEEE Trans. Evolutionary Computation	2002	Non-dominated sorting, crowding distance	General method, not portfolio-specific	Needs portfolio constraint handling
Handling multiple objectives with PSO (Coello et al.)	IEEE Trans. Evolutionary Computation	2004	External archive guides the swarm	General method, not portfolio-specific	Needs a swarm vs GA comparison
PSO for constrained portfolios (Cura; Zhu et al.; Deng et al.)	Nonlinear Anal. RWA; Expert Syst. Appl.	2009 to 2012	PSO with cardinality and other constraints	One scalarised objective	No three-objective front with CVaR
A portfolio model with three objectives and discrete variables (Anagnostopoulos, Mamanis)	Computers & Operations Research	2010	Tri-objective MOEAs: risk, return, number of securities	Third objective is holding count, not tail risk	Swarm, CVaR and held-out tests added here

Requirement Analysis and Specification

12

functional requirements

7

non-functional requirements

0

INR software cost, all open source

Software stack

Python 3.13 | NumPy | pandas | Matplotlib | SciPy | pytest | Browser canvas

Default run: 100 individuals, 150 iterations, 20 assets finishes in about 1.5 seconds

Design: architecture



Design: methodology

One repair operator, shared by both algorithms

Clip negatives to zero → Keep the K largest → Drop tiny weights → Add assets if the cap needs more → Cap at 30% and redistribute → Normalise to sum 1

Six experiments

E1 — Front quality
10 seeds against random search

E2 — Holding limits
K = none, 10, 8 and 5

E3 — Number of assets
10 to 60 assets

E4 — Search budget
5 to 200 iterations

E5 — Held-out test
30 simulated markets

E6 — Operator ablation
with and without

Seeded factor-model markets with fat tails and a crisis period; 70% training, 30% held-out

Implementation: the two optimisers

MOPSO (particle swarm)

- Particles start as feasible portfolios
- Leader chosen from the archive by crowding
- Velocity update, then repair
- Archive keeps the non-dominated set

100 particles | inertia 0.9 to 0.4 | c1 = c2 = 1.5

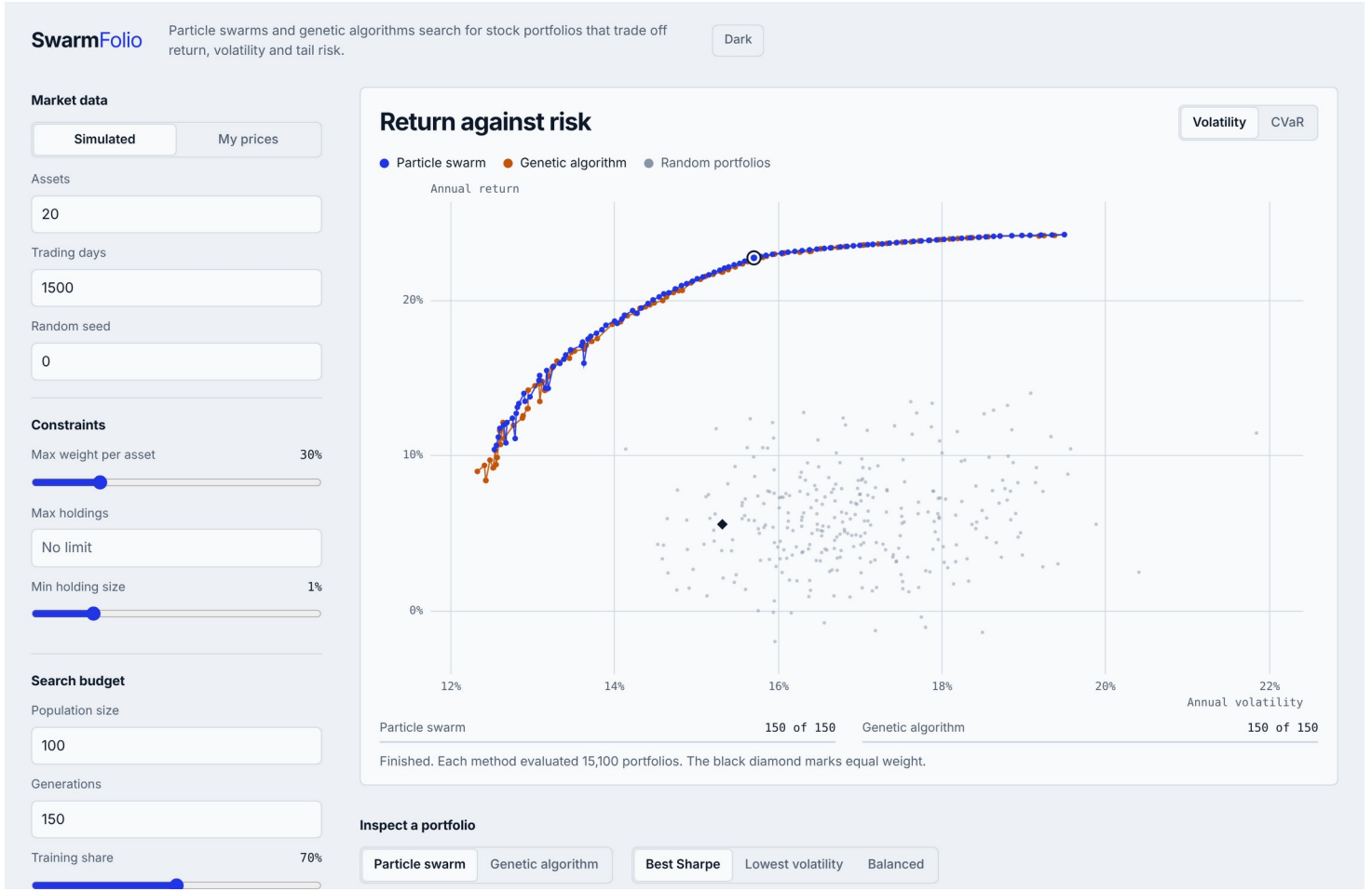
NSGA-II (genetic algorithm)

- Population of feasible portfolios
- Tournament by rank and crowding
- SBX crossover, mutation and swap
- Parents and children: keep the best

100 individuals | SBX index 15 | swap mutation 0.2

Same budget for both: 15,100 objective evaluations per run

Implementation: the workbench



Watch the search

Both populations converge on their fronts, live

Inspect any portfolio

Click a point: weights, training and held-out results

Bring your own prices

Upload a CSV, set constraints, press Run

Testing and Validation

14

automated tests pass

19

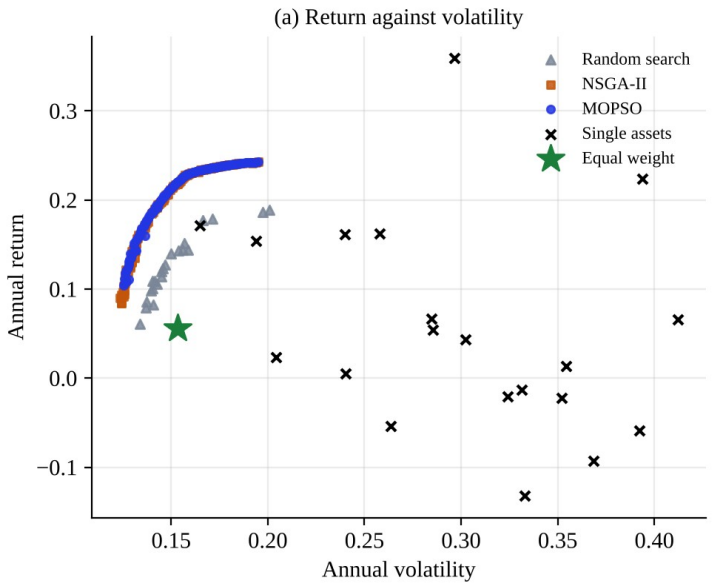
system test cases: server and browser

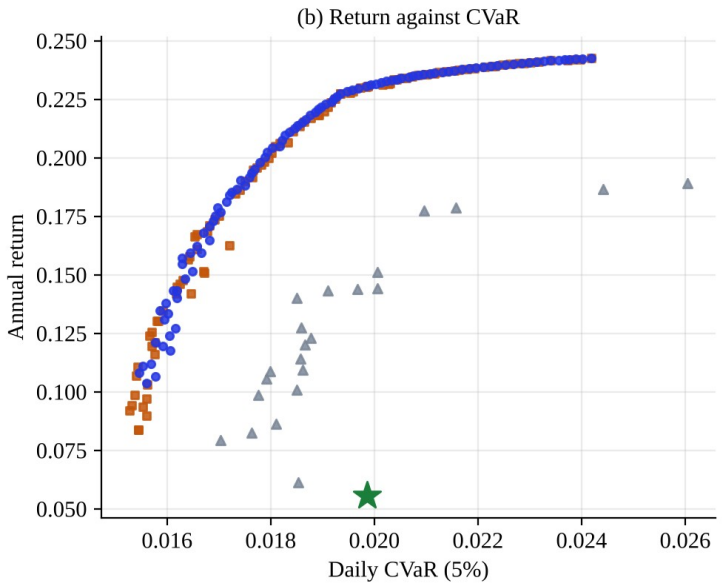
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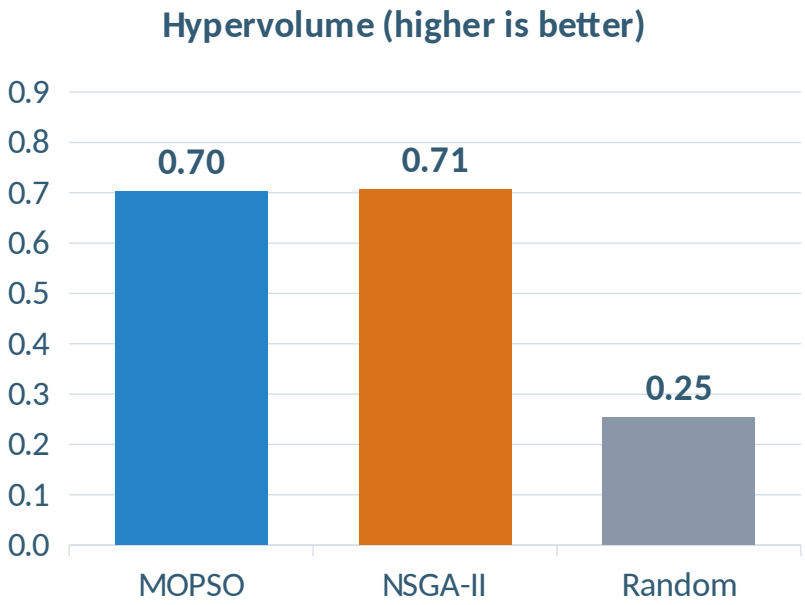
constraint violations in every experiment run

Component	Algorithm	System
Repair, objectives, Pareto maths	Feasible, non-dominated, reproducible, beats random	Server jobs, error paths, browser, phone, dark mode

Results: front quality







100 vs 20

non-dominated portfolios per run: optimisers vs random

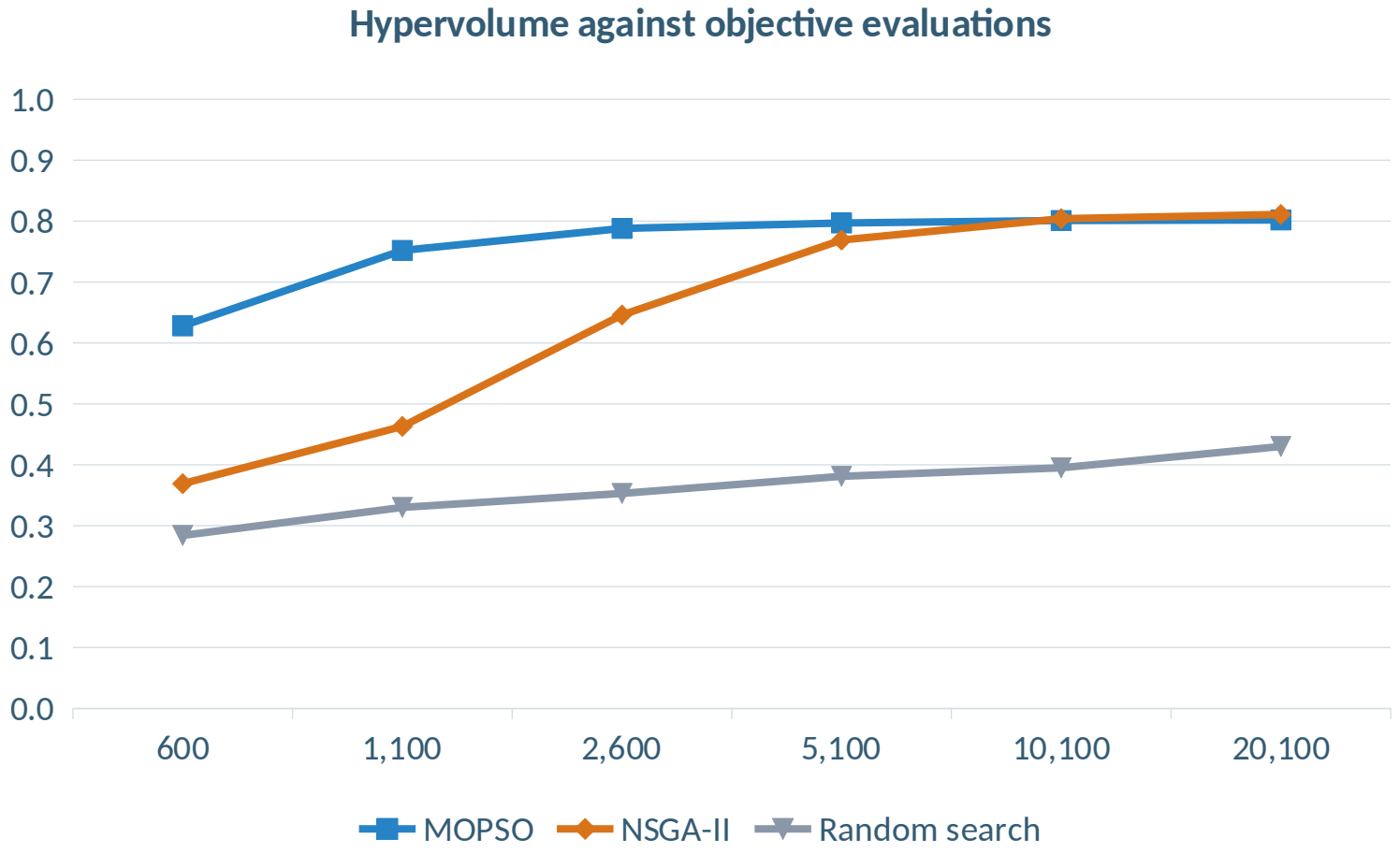
p = 0.002

both optimisers vs random, paired test over 10 seeds

0.70 vs 0.71

swarm vs genetic: a statistical tie in practice

Results: search budget



Up to 50 iterations

Swarm ahead in 5 of 5 seeds

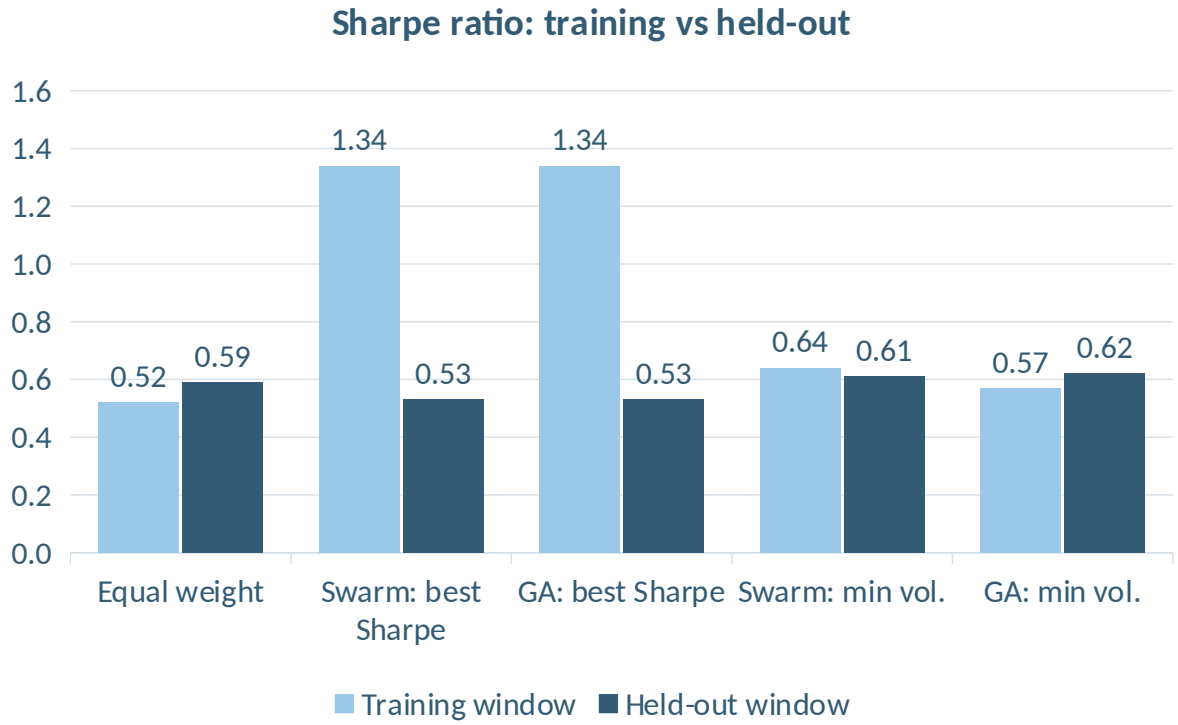
From 100 iterations

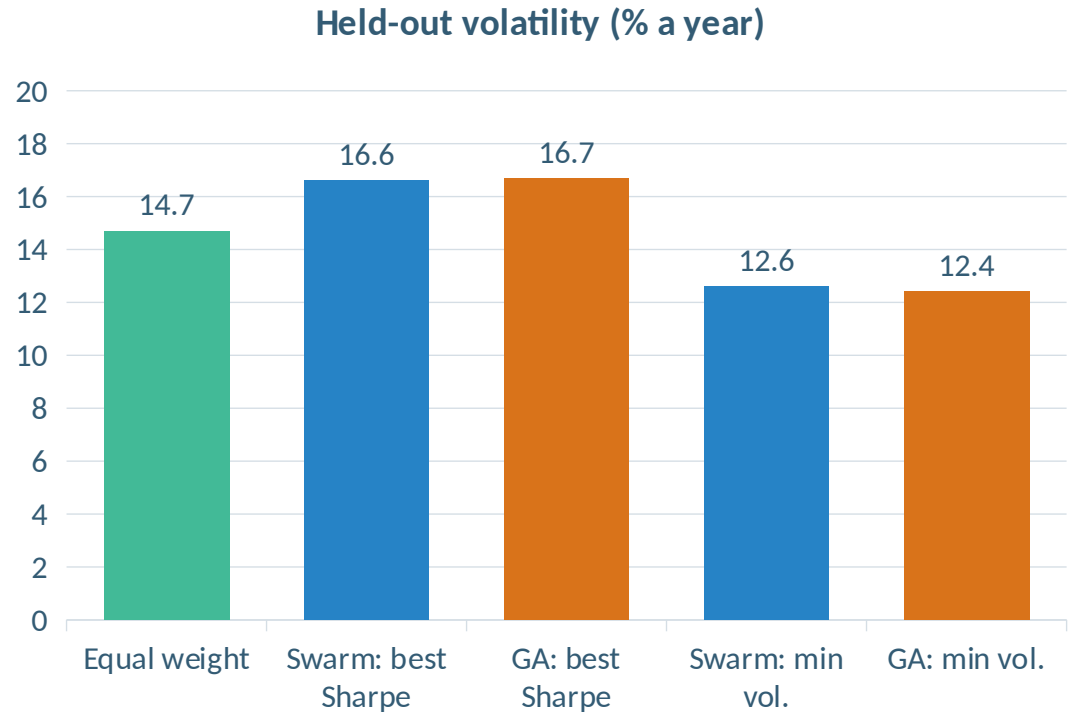
Genetic algorithm ahead in 5 of 5 seeds

Random search

Stays far behind at every budget

Results: held-out performance

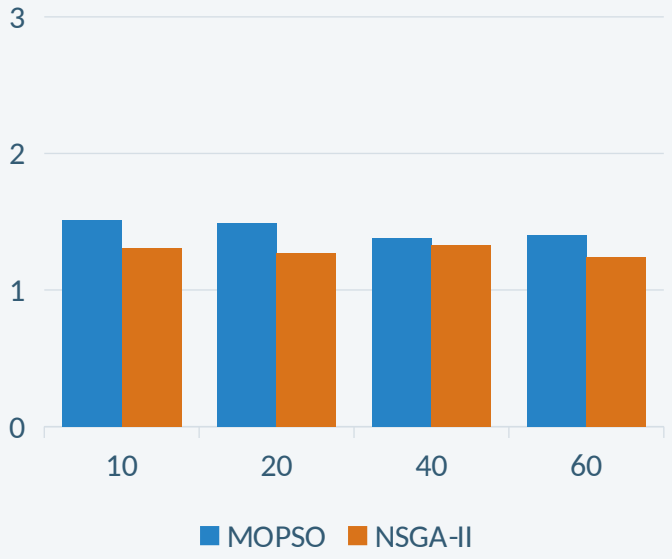




Return does not transfer to new data. Risk does: lower held-out volatility in 30 of 30 markets

Results: robustness checks

Scale



Seconds per run, 10 to 60 assets
about 1.5 s whatever the size

Constraints

0

violations of the cap or the holding limit, for limits of none, 10, 8 and 5 holdings

Operator ablation

No gain

No turbulence: p = 0.64

No swap mutation: p = 0.55

Neither changes front quality at these settings

Conclusion

1

Both optimisers work

Far better fronts than random search; every constraint respected

2

Return does not transfer

Best-Sharpe picks: 1.34 in training, 0.53 held out; equal weight 0.59

3

Risk does

Minimum-volatility picks were calmer than equal weight in 30 of 30 markets

Future scope

- Real market data
- Shrinkage estimators
- Trading costs as an objective
- Hosted classroom version

Program Outcomes Attained

3 High | 2 Medium | 1 Low

PO1 Engineering knowledge **Level 3**	**PO2** Problem analysis **Level 3**	**PO3** Design of solutions **Level 2**	**PO4** Investigation of problems **Level 3**	**PO5** Modern tool usage **Level 3**	**PO6** Engineer and the world **Level 1**	**PO7** Ethics **Level 2**
PO8 Teamwork **Level 3**	**PO9** Communication **Level 2**	**PO10** Project management **Level 2**	**PO11** Life-long learning **Level 2**	**PSO1** Design AI systems **Level 3**	**PSO2** Decisions with AI and data **Level 2**	**PSO3** End-to-end systems **Level 3**

Each level is justified in the project report (attainment sheet).